Sonit Nitin Patil

bm23btech11023@iith.ac.in | sonitnitin@gmail.com | +91 8788280330 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.30
Poddar Brio International School, Badlapur	Class XII (CBSE)	2023	96.20%
Swami Vivekananda International School, Gorai	Class X (ICSE)	2021	98.00%

EXPERIENCE

Mitsubishi Electric, Information Technology R&D Centre | Kanagawa, Japan

Research Intern

- Engineered a spiking neural network (SNN) encoding for electroencephalography (EEG) using adaptive leaky integrate and fire (ALIF) neurons in Brian2, optimised through time constants, reset potentials and injection currents.
- Validated the new encoding across convolutional SNN and liquid state machine (LSM) architectures using surrogate gradient descent and spike-timing dependent plasticity (STDP) learning, benchmarked against delta modulation encoding.
- **Achieved a 98.5% spike sparsity, 12% lower accuracy loss, and 4% higher F1** compared to delta modulation encoding **utilising only 2 EEG channels out of 22** in a motor imagery task using the standard BCI-IV 2a dataset.

Mitsubishi Electric, Advanced Technology R&D Centre | Hyogo, Japan

Research Intern

- Designed a microservices based system **integrating visual language models (VLM) and augmented reality (AR)** based control of a quadruped robot for teleoperation and autonomous control.
- **Developed a spatial computing pipeline for proprietary 3D technology system** by implementing point cloud registration, processing and immersive reconstruction on apple vision pro for real time room navigation.
- Formulated a navigation methodology leveraging VLMs to achieve zero-shot navigation without relying on SLAM.

PROJECTS

Potential (Custom SNN Library) | Self Project

- Implemented latency and poisson encoding for SNN in C++, along with a standard LIF neuron whose **parameters are computed through differential equation solved in Julia runtime within C++**.
- Identifying data structure based handling, extending neurons list with Izhikevich and Hodgkin-Huxley, and learning to transfer the C++ library to python using CMake and PyBind for extensive support.

VisProc | Self Project

- Defined the neural pathway of image understanding from the Retinal Ganglion Cells (RGCs) of the eye to the V1-V5 visual cortex regions and how semantic segmentation is the core of building edges and thus complex images in a Human Brain.
- Implemented a hardcoded classification model to accept a 2 edges image (4x4) and **identify the probability of neuron associated** with each of those edges and also the subsequent combination of edges.
- Contrasted two FCNN models with 3 layers each, one whose channels are chosen as powers of 2 (Usual NN) and other whose channels are chosen according to ideal number of neurons required to be in that layer physiologically (This NN), and found that This NN had **65% lesser inference time** for similar accuracy and F1.

3D Mapping Technology | Mitsubishi Electric Hackathon (20 Days)

- Accomplished a 60% semantic segmentation accuracy by creating a custom dynamic graph convolutional neural network (DGCNN) with chamfer ball trees KNN, and won the "Best Semantic Segmentation Model".
- Programmed a generative circuit model using Q-Learning for predicting connections among various electro-mechanical components and simulated it on a custom NextJS+threeJS website using CAD modelled components.

- Built a custom video to point cloud converter using MiDaS and Open3D to create a point cloud mapping without LiDAR.

SKILLS

AI/ML: PyTorch, NumPy, Pandas, Brian2, Matplotlib, Signal Processing

Programming Languages & Frameworks: Python, C, C++, TypeScript, NextJS

CERTIFICATIONS

Full-Stack Software Development Specialisation | IBM, Coursera

Gained practical understanding of system design, data driven development, microservices architecture and how end to end software pipelines are developed in production stage.

AWARDS & RECOGNITION

Mitsubishi Electric Hackathon

1. Won the best semantic segmentation model award with a 100,000 yen cash prize.
2. Won the most creative idea award with a 50,000 yen cash prize.

Academic Paper

Published a preprint (non peer reviewed, open source) on the subject "Collatz Conjecture". Currently has 1 citation, 700+ Views and 130+ Downloads.